
Linear Statistical Models

B. Statistical Data Science 2nd Year Indian Statistical Institute

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Revision 5 – Random Vectors

Exercise 1. In this exercise, we study mean, variance and covariance of random vectors. In the following, $\mathbf{X}$ and $\mathbf{Y}$ are random vectors in $\mathbb{R}^p$ and $\mathbb{R}^q$, respectively.

For a random vector $\mathbf{X} = (X_1, \dots, X_p)^\top$, we define $\mathbb{E}(\mathbf{X})$ to be the vector whose components are the expectations of the components of $\mathbf{X}$: $\mathbb{E}(\mathbf{X}) = (\mathbb{E}X_1, \dots, \mathbb{E}X_p)^\top$. The expectation exists if and only if $\mathbb{E}X_i$ exists for each $i = 1, \dots, p$. We can similarly define the expectation of a matrix valued random variable $\mathbf{X}$ componentwise: If $\mathbf{X} = ((X_{ij}))$, then $\mathbb{E}\mathbf{X} = ((\mathbb{E}X_{ij}))$.

(a) Check that $\mathbb{E}\mathbf{X}$ exists if and only if $\mathbb{E}\|\mathbf{X}\|_1 < \infty$.

Note: Here, $\|\cdot\|_1$ is the ℓ_1 -norm on $\mathbb{R}^p$, defined as $\|\mathbf{v}\|_1 = \sum_{i=1}^p |v_i|$.

(b) Show that if $\mathbb{E}\mathbf{X}$ exists, then $\mathbb{E}(\mathbf{A}\mathbf{X})$ exists for any non-random matrix $\mathbf{A}$ of compatible size. In this case, $\mathbb{E}(\mathbf{A}\mathbf{X}) = \mathbf{A}\mathbb{E}(\mathbf{X})$.

The variance of $\mathbf{X}$ is defined to be the matrix

$$\text{Var}(\mathbf{X}) = \begin{pmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_p) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_p) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, X_1) & \text{Cov}(X_p, X_2) & \cdots & \text{Var}(X_p) \end{pmatrix} = ((\text{Cov}(X_i, X_j))).$$

It exists if and only if $\text{Var}(X_i) < \infty$ for all $i = 1, \dots, p$.

(b) Verify that $\text{Var}(\mathbf{X})$ exists if and only if $\mathbb{E}\|\mathbf{X}\|^2 < \infty$.

(c) Show that we can write $\text{Var}(\mathbf{X}) = \mathbb{E}\{(\mathbf{X} - \mathbb{E}\mathbf{X})(\mathbf{X} - \mathbb{E}\mathbf{X})^\top\}$.

(d) Check that $\text{Var}(\mathbf{X})$ is a p.s.d. matrix. When is it p.d.?

(e) Show that if $\text{Var}(\mathbf{X})$ exists, then so does $\text{Var}(\mathbf{A}\mathbf{X})$ for any non-random matrix $\mathbf{A}$ of compatible size. In this case, $\text{Var}(\mathbf{A}\mathbf{X}) = \mathbf{A}\text{Var}(\mathbf{X})\mathbf{A}^\top$.

For two random vectors $\mathbf{X} = (X_1, \dots, X_p)^\top$ and $\mathbf{Y} = (Y_1, \dots, Y_q)^\top$ (p and q can be different), we define their covariance to be the matrix containing all the componentwise covariances:

$$\text{Cov}(\mathbf{X}, \mathbf{Y}) = \begin{pmatrix} \text{Cov}(X_1, Y_1) & \text{Cov}(X_1, Y_2) & \cdots & \text{Cov}(X_1, Y_q) \\ \text{Cov}(X_2, Y_1) & \text{Cov}(X_2, Y_2) & \cdots & \text{Cov}(X_2, Y_q) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_p, Y_1) & \text{Cov}(X_p, Y_2) & \cdots & \text{Cov}(X_p, Y_q) \end{pmatrix} = ((\text{Cov}(X_i, Y_j))).$$

It exists if both $\text{Var}(\mathbf{X})$ and $\text{Var}(\mathbf{Y})$ exist. We say that $\mathbf{X}$ and $\mathbf{Y}$ are uncorrelated if $\text{Cov}(\mathbf{X}, \mathbf{Y}) = \mathbf{O}$. If $\mathbf{X}$ and $\mathbf{Y}$ are jointly normally distributed (i.e., the big vector $(\mathbf{X}^\top, \mathbf{Y}^\top)^\top$ is multivariate normal), then $\mathbf{X}$ and $\mathbf{Y}$ are uncorrelated if and only if they are independent.

- (f) Show that $\text{Cov}(\mathbf{X}, \mathbf{Y}) = \mathbb{E}\{(\mathbf{X} - \mathbb{E}\mathbf{X})(\mathbf{Y} - \mathbb{E}\mathbf{Y})^\top\}$.
- (g) Let $\mathbf{A}$ and $\mathbf{B}$ be two non-random matrices. Then, show that $\text{Cov}(\mathbf{A}\mathbf{X}, \mathbf{B}\mathbf{Y})$ exists if $\text{Cov}(\mathbf{X}, \mathbf{Y})$ exists. In this case, $\text{Cov}(\mathbf{A}\mathbf{X}, \mathbf{B}\mathbf{Y}) = \mathbf{A}\text{Cov}(\mathbf{X}, \mathbf{Y})\mathbf{B}^\top$.
- (h) Show that $\mathbf{a}^\top \mathbf{X}$ and $\mathbf{b}^\top \mathbf{X}$ are uncorrelated if and only if $\mathbf{a}^\top \text{Var}(\mathbf{X})\mathbf{b} = 0$. Similarly, $\mathbf{a}^\top \mathbf{X}$ and $\mathbf{B}\mathbf{X}$ are uncorrelated if and only if $\mathbf{a}^\top \text{Var}(\mathbf{X})\mathbf{B}^\top = \mathbf{0}^\top$. And, $\mathbf{A}\mathbf{X}$ and $\mathbf{B}\mathbf{X}$ are uncorrelated if and only if $\mathbf{A}\text{Var}(\mathbf{X})\mathbf{B}^\top = \mathbf{O}$.

Note: If $\text{Var}(\mathbf{X}) = \sigma^2 \mathbf{I}$, then the conditions boil down to $\mathbf{a}^\top \mathbf{b} = 0$ (i.e., $\mathbf{a} \perp \mathbf{b}$), $\mathbf{B}\mathbf{a} = \mathbf{0}$ (i.e., $\mathbf{a} \perp \mathcal{R}(\mathbf{B})$), and $\mathbf{A}\mathbf{B}^\top = \mathbf{O}$ (i.e., $\mathcal{R}(\mathbf{A}) \perp \mathcal{R}(\mathbf{B})$). If $\mathbf{X}$ is multivariate normal with variance $\sigma^2 \mathbf{I}$, then these conditions imply independence between $\mathbf{a}^\top \mathbf{X}$ & $\mathbf{b}^\top \mathbf{X}$, $\mathbf{a}^\top \mathbf{X}$ & $\mathbf{B}\mathbf{X}$, and $\mathbf{A}\mathbf{X}$ & $\mathbf{B}\mathbf{X}$, respectively.